



xScape SpaceWire

Interface Control Document

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Author(s):	J Vos, G Dreijer
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Document History

Revision	Date	Details
1	2021-04-07	First Release
2	2021-06-22	Updated Section 3.4.1 to note Control and Data Host Logical addresses must be two different addresses and cannot be the same, and that Data Host Logical Address can be set to 0x00 to disable it. Updated Section 3 and added information on Simera Sense Standard Protocol. Updated Section 3 and removed DSID from CCSDS Protocol Definition.
3	2024-02-05	Corrected parameter number for data interface parameter in Table 3-2 Updated section 3.1 – Removed pin assignments as the details are specified in the product’s Interface Control Document. Updated Section 3.3 to include the optional Secondary Header Ancillary Bytes. Updated section 3.4.2.1 – Added more detail.

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List of Abbreviations

Abbreviation	Description
I/O	Input/Output
LVDS	Low Voltage Differential Signalling
Mbps	Megabits per second
RX	Receive
TX	Transmit

1 Introduction

1.1 Identification

Item Description: xScape SpaceWire

1.2 Intended Use

This document describes the xScape optional SpaceWire interface.

1.3 Context and Summary

The xScape range of products support various options for control and/or data interfaces, including SpaceWire. The SpaceWire interface provides a control interface as well as a high-speed data transfer interface. A single SpaceWire interface can perform both control and data transfer functions at the same time.

2 Referenced Documents

Table 2-1 lists documents that are referenced in this document. In the event of conflict between the contents of the reference documents and this document, this document shall take precedence.

Table 2-1: Referenced Documents

Ref. #	Reference
[1]	ECSS-E-ST-50-12C - SpaceWire – Links, nodes, routers and networks
[2]	CCSDS 133.0.B-1 - Space Packet Protocol, Recommended Standard
[3]	051715 - xScape Control and Data Interface Control Document

For undated references, the latest released version of the reference document applies. For dated references, subsequent versions of the document do not apply. It is best practice to always refer to the latest released version. Unless otherwise stated, web links referenced above were last accessed at the release date of the current version of this document.

3 Interface Description

The imager provides a high-speed SpaceWire interface that is routed to connector P5 by default. The SpaceWire link conforms to [1] and can operate at up to 100 Mbps in either direction. A single SpaceWire node is implemented having two services (end-points): a Data Service and a Control Service. The SpaceWire interface can be configured to communicate with one node, or two nodes concurrently, one for control and one for data. Provision is made for up to 8 SpaceWire routing bytes, independently configurable for both the control and data services. These routing bytes (including logical addresses) listed in Table 3-1 are configured using Default System Parameters, as described in section 3.4.1.

Table 3-1: SpaceWire Routing Parameters

Parameter	Size	Notes
Imager Logical Address	1 byte	Used by the Imager when receiving command and request packets
Control Host Logical Address	1 byte	Used by the Imager when sending request response packets
Control Host Routing Bytes	7 bytes	Used by the Imager when sending request response packets
Data Host Logical Address	1 byte	Used by the Imager when sending data packets
Data Host Routing Bytes	7 bytes	Used by the Imager when sending data packets

Two standard protocol options are available. The Simera Sense Standard Protocol follows a minimalistic approach, while the CCSDS packet protocol follows the recommendation as per [2]. The CCSDS packet protocol is available with and without Secondary Header option. The control service protocol cannot be changed on-the-fly and is a build option. The data service protocol can however be changed using Default System Parameters, as described in section 3.4.1.

3.1 Physical Interface

The SpaceWire physical interface is specified in the product's Interface Control Document.

3.2 Simera Sense Standard Protocol

3.2.1 Control Transactions

For both command and request transactions, SpaceWire transfers use SpaceWire packets. Each control transfer packet has up to 7 routing bytes followed by the logical address and a variable length control packet. The Imager standard command or request packet (refer to [3]) is inserted in its original, little endian, format.

Command Transactions

Command transactions require the SpaceWire host to send a command transfer packet, as shown in Figure 3-1. The logical address must match that of the Imager Default Parameter ID 0x41, described in section 3.4.1.

Request Transactions

Request transactions are carried out by two transfers. Firstly, the SpaceWire host sends a request transfer packet, as shown in Figure 3-2. The routing bytes must be set up according to the SpaceWire network map, with the logical address matching that of the Imager. The imager then responds by sending a request response transfer packet, as shown in Figure 3-3, to the host. The Imager will set the routing bytes (including host logical address) according to the Default System Parameters for the SpaceWire Host, described in section 3.4.1.

3.2.2 Data Transactions

SpaceWire data transactions consist of SpaceWire packets that are sent from the Imager to the host. Each packet has up to 8 routing bytes followed by a variable length data packet in little endian format. The SpaceWire data transfer packet is shown in Figure 3-4.

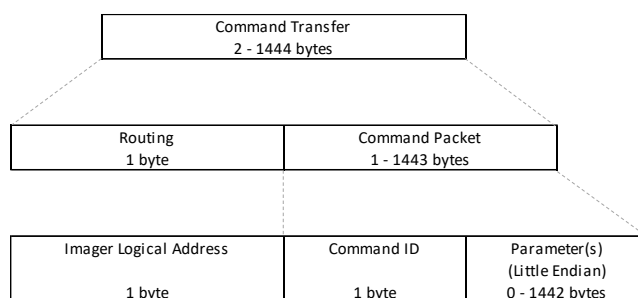


Figure 3-1: SpaceWire Command Transaction – Simera Sense Standard Protocol

Notes:

- a) The “Imager Logical Address” is the logical address of the Imager’s SpaceWire Control Service.

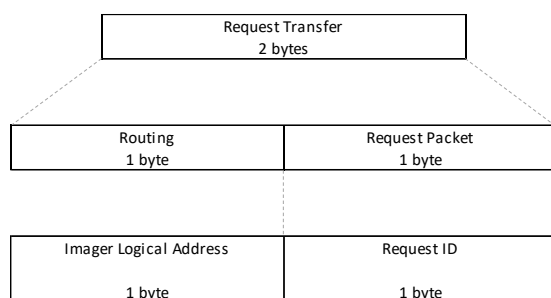


Figure 3-2: SpaceWire Request Transfer – Simera Sense Standard Protocol

Notes:

- a) The “Imager Logical Address” is the logical address of the Imager’s SpaceWire Control Service.

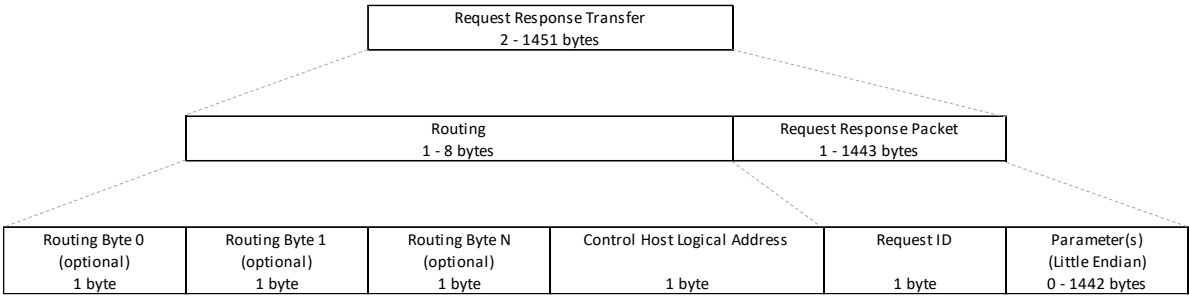


Figure 3-3: SpaceWire Request Response Transfer – Simera Sense Standard Protocol

Notes:

- a) The “Control Host Logical Address” is the logical address of the device the packet is sent to.

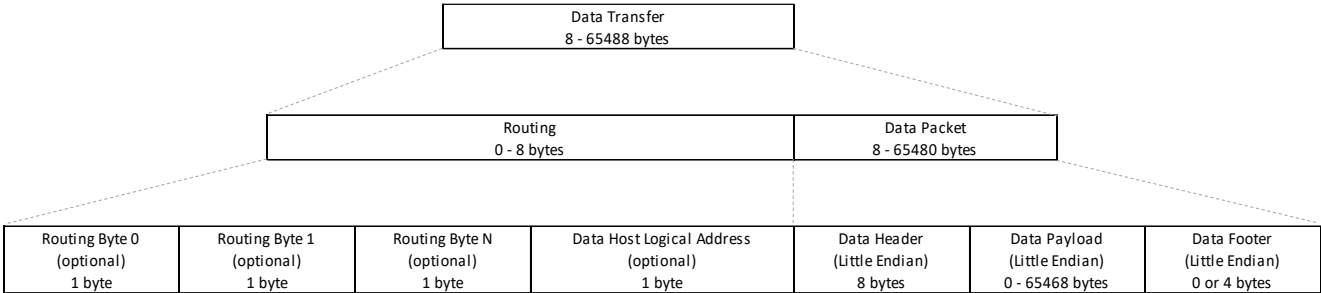


Figure 3-4: SpaceWire Data Transaction – Simera Sense Standard Protocol

Notes:

- a) The Routing Bytes are optional.
- b) The “Data Host Logical Address” is the logical address of the device the packet is sent to. (optional)

3.3 CCSDS Packet Protocol

3.3.1 Control Transactions

For both command and request transactions, SpaceWire transfers use SpaceWire packets. Each control transfer packet has up to 7 routing bytes followed by the logical address and a variable length CCSDS packet. The CCSDS packet's primary and secondary headers are represented in big endian format, while the CCSDS packet's data field is in little endian format. The Imager standard command or request packet (refer to [3]) is inserted into the CCSDS packet data field in its original, little endian, format. The CCSDS packet secondary header is optional and consists of a 9-byte Time Code field and an optional 2-byte Ancillary Data field.

3.3.1.1 Command Transactions

Command transactions require the SpaceWire host to send a command transfer packet, as shown in Figure 3-5. Fields marked "N/A" are not applicable (don't care) to the transaction and their contents are ignored. The logical address must match that of the Imager.

3.3.1.2 Request Transactions

Request transactions are carried out by two transfers. Firstly, the SpaceWire host sends a request transfer packet, as shown in Figure 3-6. Fields marked "N/A" are not applicable (don't care) to the transaction. The routing bytes must be setup according to the SpaceWire network map, with the logical address matching that of the Imager. The imager then responds by sending a request response transfer packet, as shown in Figure 3-7, to the host. Fields marked "N/A" are not applicable (don't care) to the transaction. The Imager will set the routing bytes (including host logical address) according to the Default System Parameters for the SpaceWire Host, described in section 3.4.1. The value for the request response transfer APID field can be specified and is a build option. If enabled, the value for the Secondary Header Ancillary Data field can also be specified as a build option.

3.3.2 Data Transactions

SpaceWire data transactions consist of SpaceWire packets that are sent from the Imager to the host. Each packet has up to 8 routing bytes followed by a variable length CCSDS packet (ranging from 14 bytes up to 65497 bytes). The CCSDS packet's primary and secondary headers are represented in big endian format, while the CCSDS packet's data field is in little endian format. The SpaceWire data transfer packet is shown in Figure 3-8. Fields marked "N/A" are not applicable (don't care) to the transaction. The value for the APID field can be specified and is a build option. If enabled, the value for the Secondary Header Ancillary Data field can also be specified as a build option.

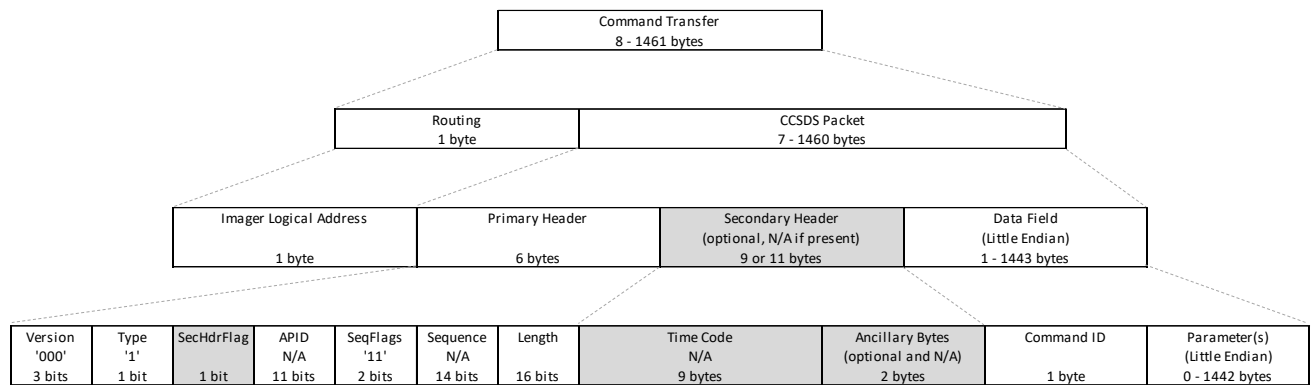


Figure 3-5: SpaceWire Command Transaction – CCSDS Packet Protocol

Notes:

- The “Imager Logical Address” is the logical address of the Imager’s SpaceWire Control Service.
- When the Secondary Header Flag (“SecHdrFlag”) is set, it signifies the presence of the Secondary Header. When present, the contents of the Secondary Header is not used by the Imager and can be set to any value. The presence of the Secondary Header Ancillary Bytes can only be enabled when the Secondary Header Time Code is also enabled.
- Primary Header “APID” field is not used by the Imager and can be set to any value.
- Primary Header “Sequence” field is not used by the Imager and can be set to any value.
- The Primary Header “Length” field indicates the number of remaining bytes in the transfer minus 1 byte, as per [2].

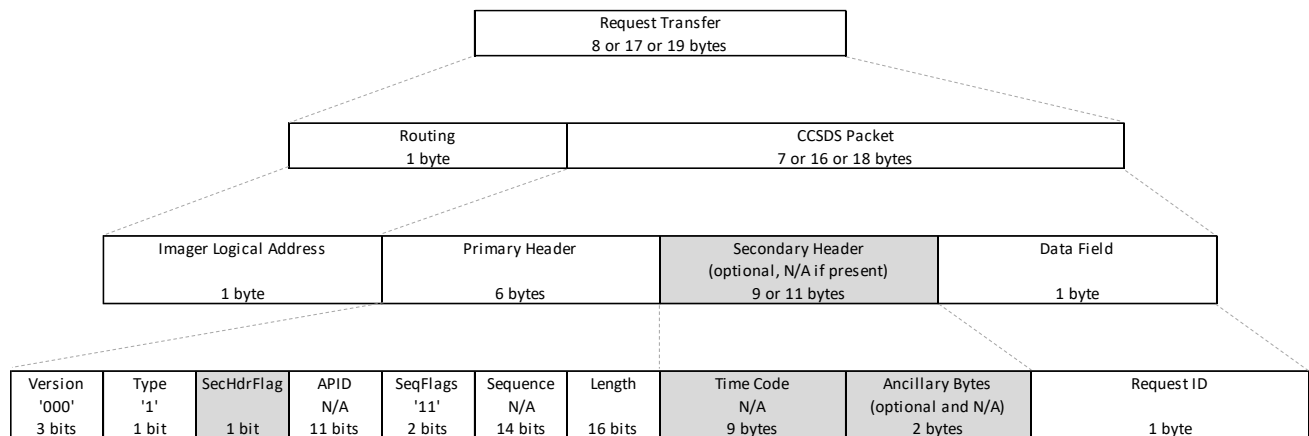


Figure 3-6: SpaceWire Request Transfer – CCSDS Packet Protocol

Notes:

- The “Imager Logical Address” is the logical address of the Imager’s SpaceWire Control Service.
- When the Secondary Header Flag (“SecHdrFlag”) is set, it signifies the presence of the Secondary Header. When present, the contents of the Secondary Header is not used by the Imager and can be set to any value. The presence of the Secondary Header Ancillary Bytes can only be enabled when the Secondary Header Time Code is also enabled.
- Primary Header “APID” field is not used by the Imager and can be set to any value.
- Primary Header “Sequence” field is not used by the Imager and can be set to any value.
- The Primary Header “Length” field indicates the number of remaining bytes in the transfer minus 1 byte, as per [2].

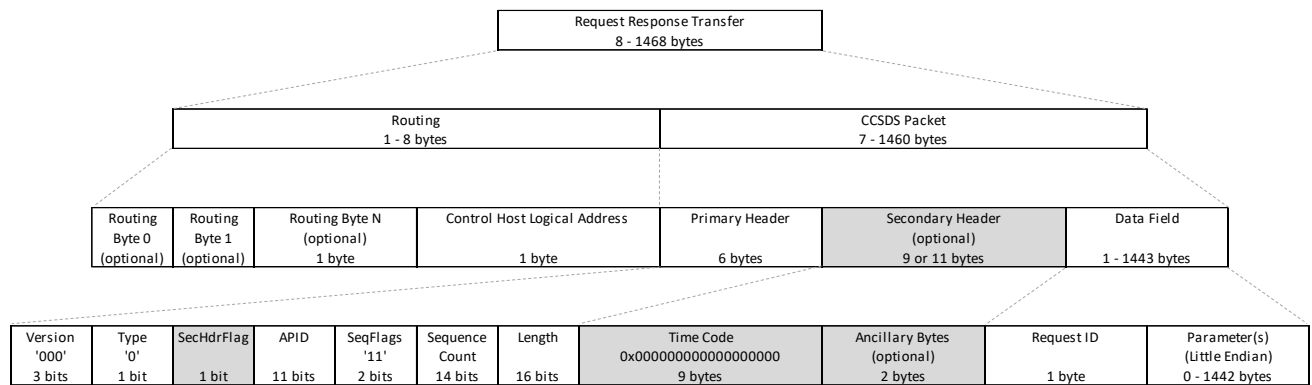


Figure 3-7: SpaceWire Request Response Transfer – CCSDS Packet Protocol

Notes:

- The “Control Host Logical Address” is the logical address of the device the packet is sent to.
- When the Secondary Header Flag (“SecHdrFlag”) is set, it signifies the presence of the Secondary Header. When present, the contents of the Secondary Header Time Code is 0x0000000000000000. The value for the Secondary Header Ancillary Bytes defaults to 0x0000 when enabled but can be set to any value (a fixed value, the same for all request response packets) if desired as a build option.
- Primary Header “APID” field defaults to 0x001. It can however be set to any value (a fixed value, the same for all request response packets) if desired as a build option.
- Primary Header “Sequence” field is incremented for each packet transmitted. Its value is reset to 0x0000 at Imager reset or power cycle.
- The 16-bit Primary Header “Length” field indicates the number of remaining bytes in the transfer minus 1 byte, as per [2].

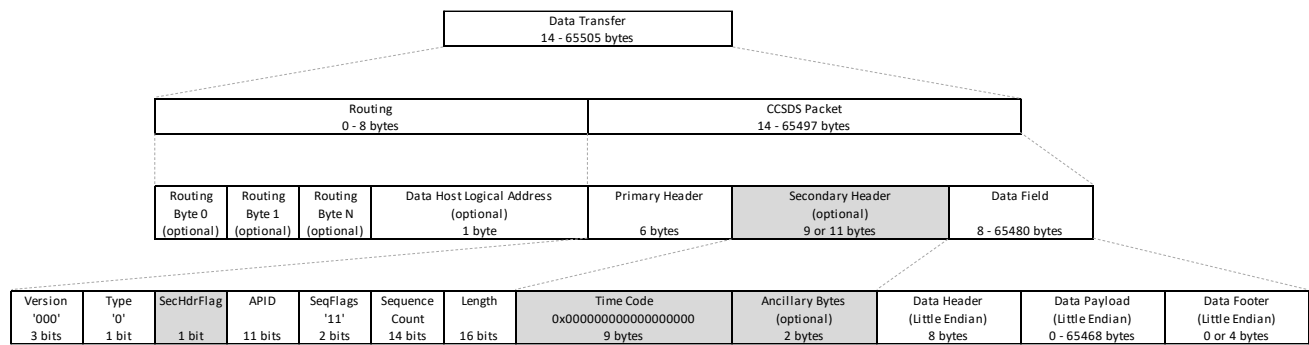


Figure 3-8: SpaceWire Data Transaction – CCSDS Packet Protocol

Notes:

- The Routing Bytes are optional.
- The “Data Host Logical Address” is the logical address of the device the packet is sent to. (optional)
- When the Secondary Header Flag (“SecHdrFlag”) is set, it signifies the presence of the Secondary Header. When present, the contents of the Secondary Header Time Code is 0x0000000000000000. The value for the Secondary Header Ancillary Bytes defaults to 0x0000 when enabled but can be set to any value (a fixed value, the same for all data packets) if desired as a build option.
- Primary Header “APID” field is not used by the Imager and defaults to 0x000. It can however be set to any value (a fixed value, the same for all data packets) if desired as a build option.
- Primary Header “Sequence” field is incremented for each packet transmitted. It’s value is reset to 0x0000 at Imager reset or power cycle.
- The 16-bit Primary Header “Length” field indicates the number of remaining bytes in the transfer minus 1 byte, as per [2]

3.4 Software Interface

3.4.1 SpaceWire Default System Parameters

The inclusion of the SpaceWire Interface adds a few Default System Parameters. The default system parameters are stored in non-volatile memory and loaded at power-up or after a system reset. It is not possible to change any system parameter while operating the imager. The SET DEFAULT SYSTEM PARAMETER command may be used to save a new default value for the applicable parameter, refer to [3].

The Default System Parameters need to be configured only once and it is not necessary to modify them under normal operating conditions.

Table 3-2: Additional Default System Parameters

Parameter Number		Parameter Value	
		Type	Description
General			
0x02	Data Interface	uint8	Additional option for the SpaceWire Data Service: 3 : SpaceWire
SpaceWire			
0x40	Baud Rate	uint8	SpaceWire Interface Transmit Baud rate. 10: 10 Mbps 20: 20 Mbps 40: 40 Mbps 50: 50 Mbps 100: 100 Mbps
0x41	Imager Logical Address	uint8	The logical address of the Imager. Used when receiving packets (commands and requests).
0x42	Control Host Logical Address	uint8	The logical address of the Control Interface Host. Used when transmitting packets on the Control Interface (request responses). This address must be different to that of "Data Host Logical address". See parameter number 0x44.
0x43	Control Host Routing Bytes	uint8	Routing Byte Number: 0 to 6
		uint8	Routing Byte Value. Used when transmitting packets on the Control Interface (requests). Unused routing bytes are set to zero ('0'). Routing bytes are injected from number 0 onward until the first zero value is detected. By writing zero to Routing Byte number 0, no routing byte is injected and the first byte transmitted is the Control Host Logical Address.

0x44	Data Host Logical Address	uint8	The logical address of the Data Interface Host. Used when transmitting packets on the Data Interface. This address must be different to that of “Control Host Logical address”. See parameter number 0x42. Set this value to 0x00 to disable and not transmit this byte.
0x45	Data Host Routing Bytes	uint8	Routing Byte Number: 0 to 6
		uint8	Routing Byte Value. Used when transmitting packets on the Data Interface. Unused routing bytes are set to zero ('0'). Routing bytes are injected from number 0 onward until the first zero value is detected. By writing zero to Routing Byte number 0, no routing byte is injected and the first byte transmitted is the Data Host Logical Address (if the Data Host Logical Address parameter, ID 0x44, has non-zero value).
0x46	Data Packet Protocol	unit8	0 : Simera Sense Standard Protocol 2 : CCSDS Packet Protocol, excluding Secondary Header. SecHdrFlag is cleared ('0'). 3 : CCSDS Packet Protocol, including Secondary Header. SecHdrFlag is set ('1').
0x47	Data Payload Size Limit	uint16	Limit transaction size for data transfers. 0 : No Limit 4 to 65532 : Limit the “Data Payload” field (excluding headers and routing bytes) to the number of bytes as specified in this value. This value must be a multiple of 4.

3.4.2 SpaceWire Requests

The inclusion of the SpaceWire Interface adds the following two Requests.

Table 3-3: Additional Request Identifiers

Request ID	Packet Name	Description
0x95	SPACEWIRE LINK STATUS	SpaceWire Control/Data Link Status
0x96	SPACEWIRE PACKET DIAGNOSTICS	SpaceWire Control Interface packet information

3.4.2.1 SpaceWire Link Status

The SPACEWIRE LINK STATUS request returns the status of the SpaceWire Link (used as a Control and/or Data Interface). The Status field includes the state of the link which indicates if a connection has been established between the Imager’s SpaceWire node and another SpaceWire node.

SpaceWire Link Status														
Request ID	0x95													
Data Bytes	2													
Return Data	Type	Description												
Status	uint8	Status of the SpaceWire Link. <table><tr><th>Bit</th><th>Description</th></tr><tr><td rowspan="4">0..1</td><td>Connection State. 0 : Disabled 1 : Enabled, disconnected 2: Busy Connecting 3: Connected</td></tr><tr><td>2</td><td>Error Detected (see error flags)</td></tr><tr><td>3</td><td>Transmit Queue Empty</td></tr><tr><td>4..7</td><td>Reserved</td></tr></table>	Bit	Description	0..1	Connection State. 0 : Disabled 1 : Enabled, disconnected 2: Busy Connecting 3: Connected	2	Error Detected (see error flags)	3	Transmit Queue Empty	4..7	Reserved		
		Bit	Description											
		0..1	Connection State. 0 : Disabled 1 : Enabled, disconnected 2: Busy Connecting 3: Connected											
			2	Error Detected (see error flags)										
			3	Transmit Queue Empty										
			4..7	Reserved										
		Error Flags	uint8	SpaceWire Link error flags. If an error occurred, the appropriate bits will be set high. <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Disconnect Error</td></tr><tr><td>1</td><td>Parity error</td></tr><tr><td>2</td><td>Escape Error</td></tr><tr><td>3</td><td>Credit Error</td></tr><tr><td>4..7</td><td>Reserved</td></tr></table>	Bit	Description	0	Disconnect Error	1	Parity error	2	Escape Error	3	Credit Error
Bit	Description													
0	Disconnect Error													
1	Parity error													
2	Escape Error													
3	Credit Error													
4..7	Reserved													

3.4.2.2 SpaceWire Packet Diagnostics

The SPACEWIRE PACKET DIAGNOSTICS request returns diagnostic information about the SpaceWire Packets of the Control Interface only. This information is used to diagnose connection problems on the SpaceWire Control Interface.

SpaceWire Packet Diagnostics		
Request ID	0x96	
Data Bytes	17	
Return Data	Type	Description
Packet RX Count	uint32	A count of the total of packets received by the imager since the last reset or power-up.
Packet RX Fail	uint32	A count of the failed packets received by the imager since the last reset or power-up. Failed Packets will have an error, as indicated in the Error Flags field.
Packet RX OK	uint32	A count of the successful packets (no errors) received by the imager since the last reset or power-up.

Packet TX	uint32	A count of the total of packets transmitted by the imager since the last reset or power-up.								
Packet RX Error Flags	uint8	Accumulated errors for all packets received since this request was last issued. <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Packet too short</td></tr><tr><td>1</td><td>CCSDS Length Field too short</td></tr><tr><td>2..7</td><td>Reserved</td></tr></table> Note: Packet Error Flags are cleared when this request is received.	Bit	Description	0	Packet too short	1	CCSDS Length Field too short	2..7	Reserved
Bit	Description									
0	Packet too short									
1	CCSDS Length Field too short									
2..7	Reserved									

Contact Us

Simera Sense (Pty) Ltd

Old Paardevlei Road

Somerset West, 7130

South Africa

info@simera-sense.com

www.simera-sense.com

Simera Sense Europe BV

Technologielaan 9

3001 Leuven (Heverlee)

Belgium

info@simera-sense.com

www.simera-sense.com